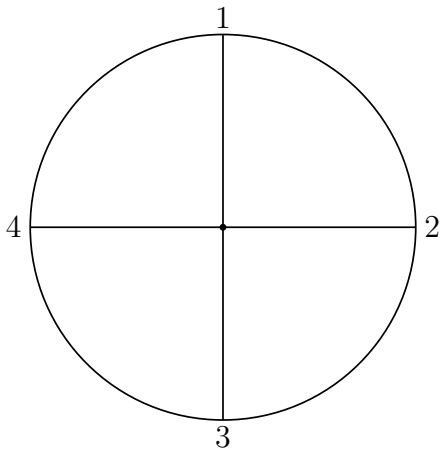
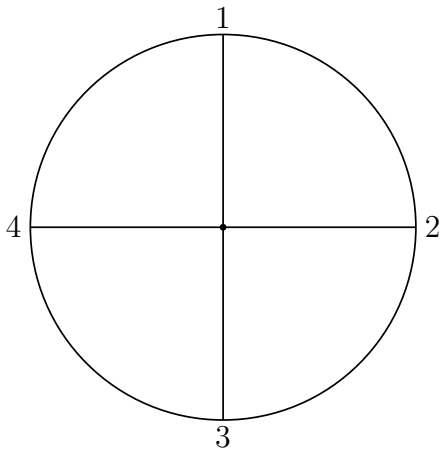
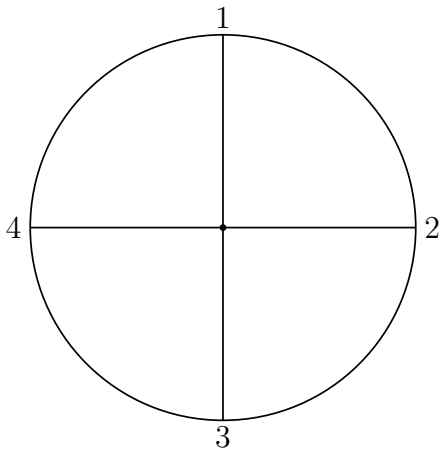
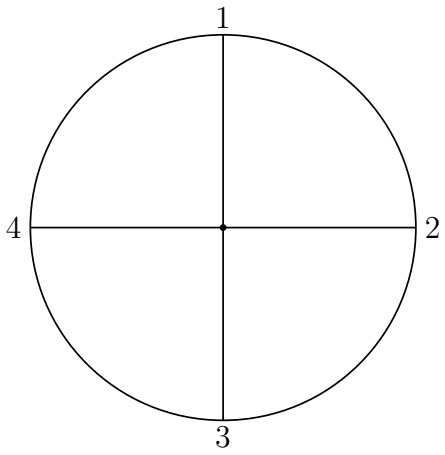
Use the marked front when you label folds. M means mountain; V means valley. Fold every ray fully flat. Keep the paper whole, with no extra creases.

Problem 1: Make the four-ray disk close flat in four different ways. Mark each fold's mountains M and valleys V.



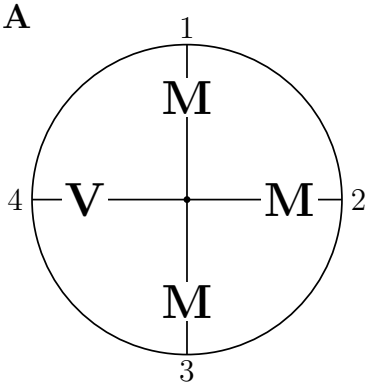




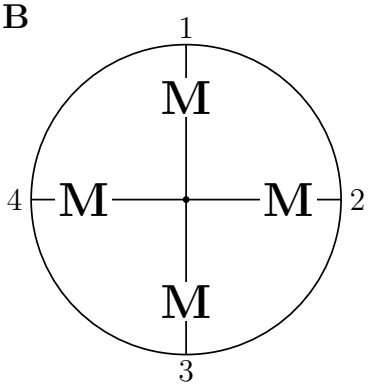


Problem 2: Which of these folds can you make with the four-ray disk? Circle each one you make.

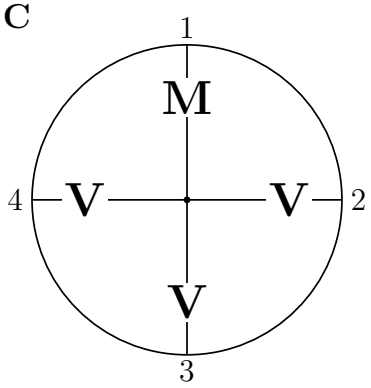
A



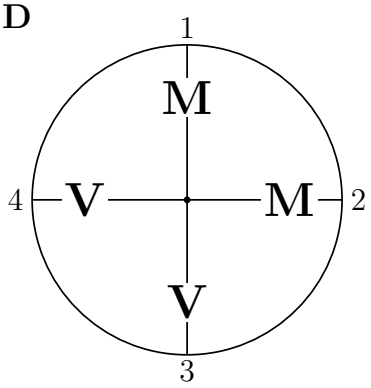
B



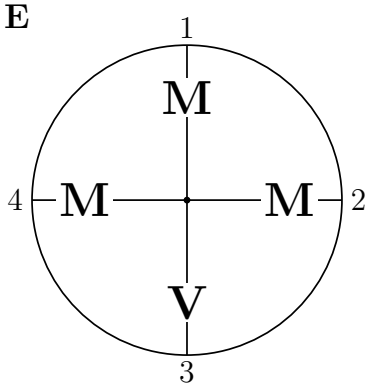
C



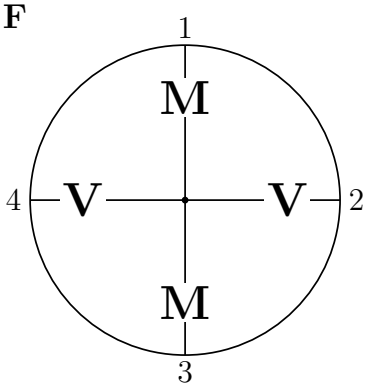
D



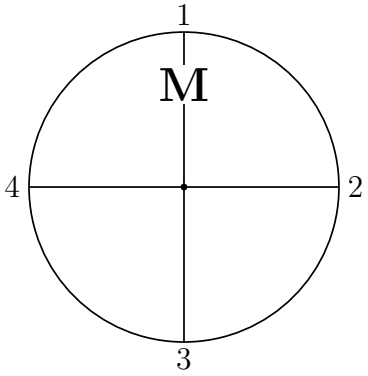
E

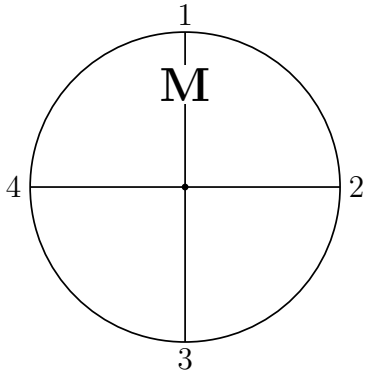


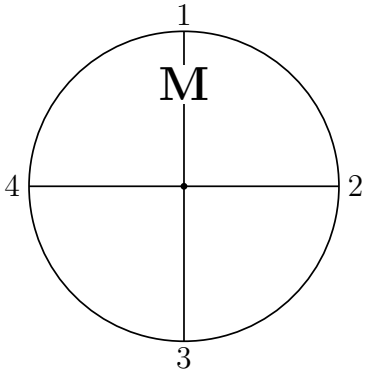
F

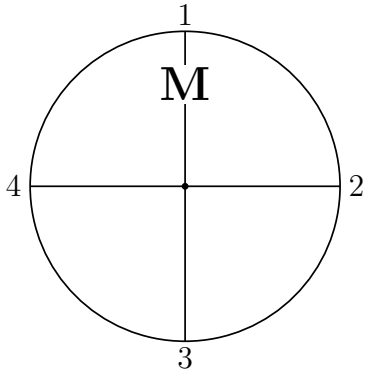


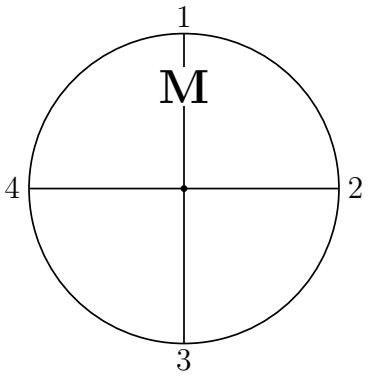
Problem 3: Ray 1 must be a mountain. Find every flat fold with that label.

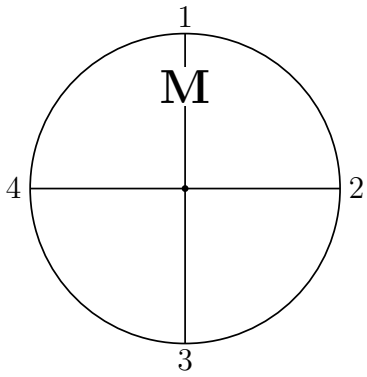






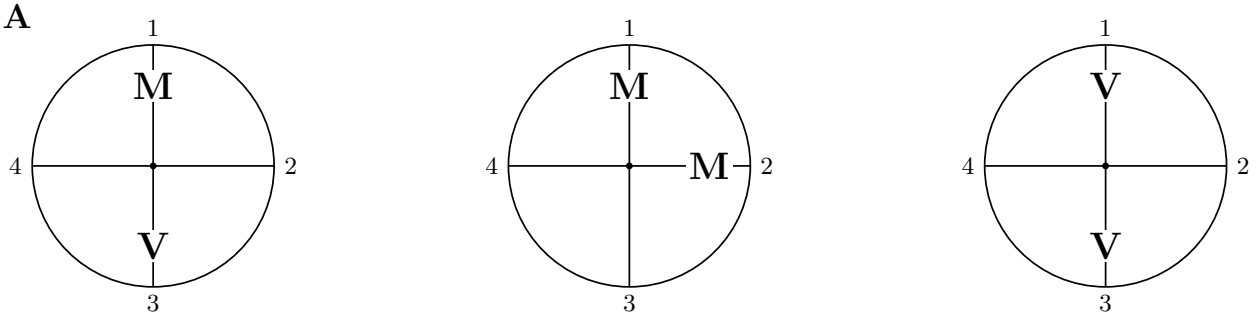






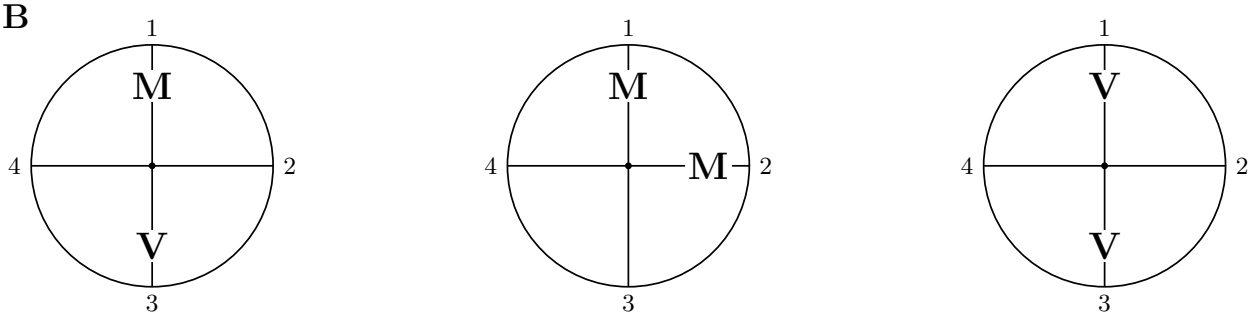
Problem 4: For each row, use its six tabs to finish the disks. Can all three disks close flat?

A



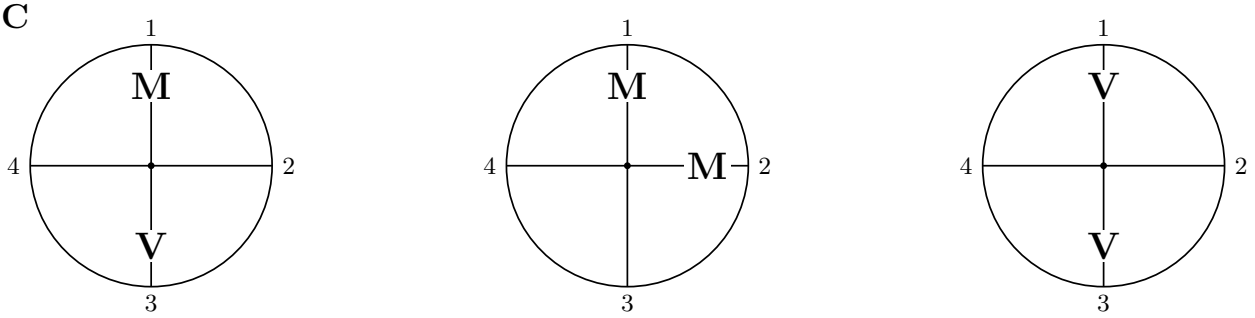
M	**M**	**V**	**V**	**V**	**V**

B



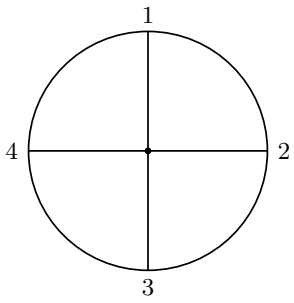
M	**M**	**M**	**V**	**V**	**V**

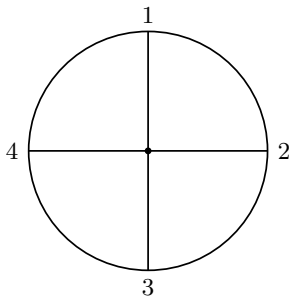
C

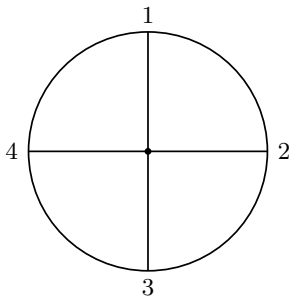


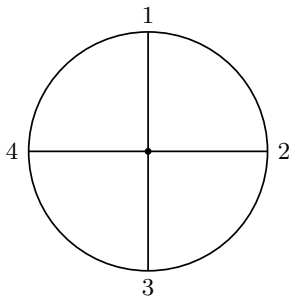
M	**M**	**M**	**M**	**V**	**V**

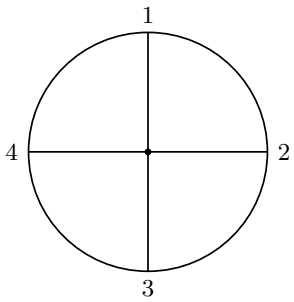
Problem 5: Find every flat fold of the four-ray disk. Keep ray 1 at the top and mark every ray M or V.

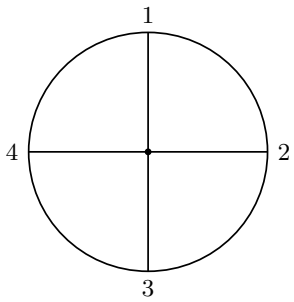


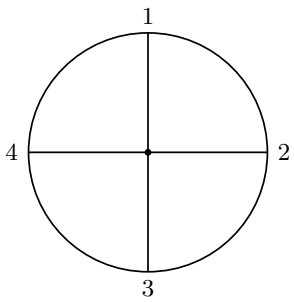


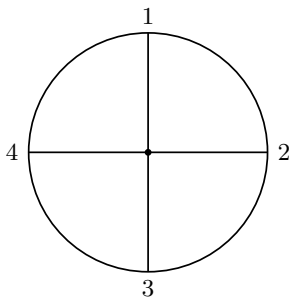


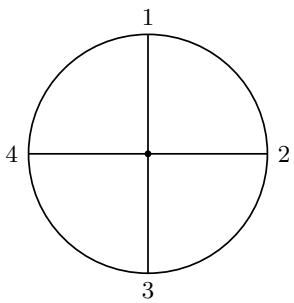


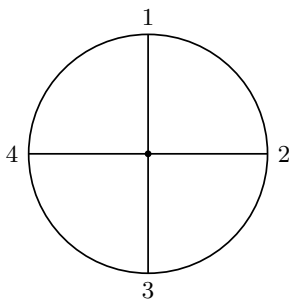






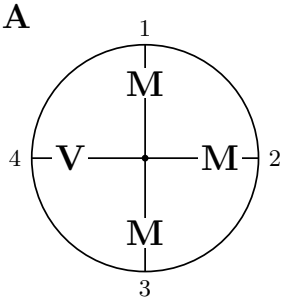




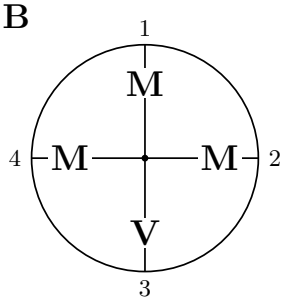


Problem 6: Visit all eight folds once, changing exactly two labels at each move. Find a route from A to H and a route from A to D.

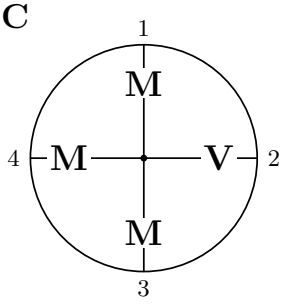
A



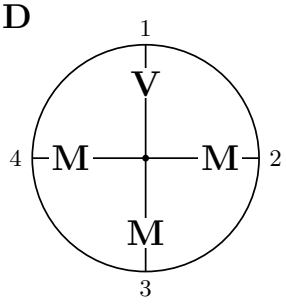
B



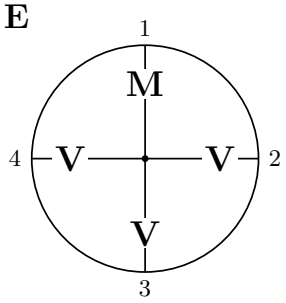
C



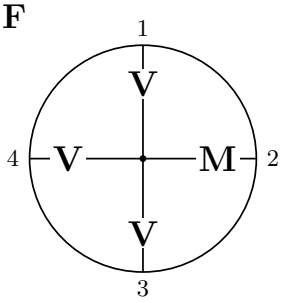
D



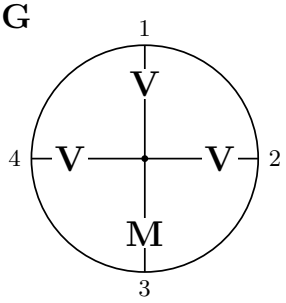
E



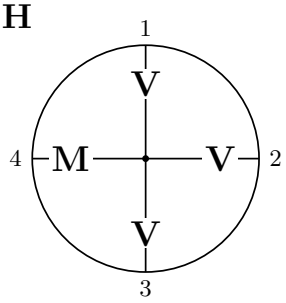
F



G



H



A ⟶ _____ ⟶ _____ ⟶ _____ ⟶ _____ ⟶ _____ ⟶ _____ ⟶ **H**

A ⟶ _____ ⟶ _____ ⟶ _____ ⟶ _____ ⟶ _____ ⟶ _____ ⟶ **D**